

An abstract graphic on the left side of the page, consisting of a complex network of black dots (nodes) connected by thin black lines (edges). The network is dense and irregular, with many small triangles and polygons formed by the connections. It appears to be a representation of a digital network or a complex system.

Cyber Attack Project

CONNIE HA

Introduction

Name: Connie Ha

Year: Sophomore

Major: Computer Science -> CST (Security Concentration)

Fun fact: I'm currently rewatching Grey's Anatomy in preparation for season 20!

Overview of the Hack

WHO was hacked?

T-Mobile USA

WHAT was stolen?

Personal data of an estimated 76.6 million T-Mobile customers were stolen in this attack according to court filings, but hackers approximated the number to more than 100 million users. The data included names, social security numbers, security pins, phone numbers, addresses, date of births, and other information unique to each mobile device of the customers.

WHERE did the attack occur?

T-Mobile's wireless data network

Overview of the Hack CONT

WHEN did the hack occur?

T-Mobile reported that the cyberattack occurred on August 15, 2021.

WHY were they targeted?

Researcher Alon Gal stated that the hackers responsible for this data breach did it to “retaliate against the US for the kidnapping and torture of John Erin Binns in Germany by the CIA and Turkish intelligence agents in 2019. We did it to harm US infrastructure.” It’s also due to the vulnerabilities found in T-Mobiles data network.

Attack Vectors Used

How were they attacked?

The hackers responsible for this attack found an opening in T-Mobile's wireless data network that was the gateway to giving them access to two of T-Mobile's customer data centers.

How did the hackers enter the network?

T-Mobile disclosed that the hackers were able to enter their network and gain access to customer data due to exploiting a vulnerable API.

Description of Cyber Tools Used

What tools did the hackers use to gain access to the network?

Vulnerable API's are usually caused by weak authentication and exposed sensitive data, which means the hackers used phishing/spear phishing and privilege escalation to gain access to the T-Mobile networks.

What flaws in security were taken advantage of by the hackers?

With the vulnerable API, the hackers took advantage of T-Mobile's outdated software, weak authentication mechanisms, and insufficient employee training/awareness.

Hacker Identification

Has the hacker been identified?

Yes, it was identified that IntelSecrets, also known as IRDev and Vortex hacked into T-Mobile's network.

Have any charges been filed against the hacker?

No charges were filed against the hacker.

Repercussions of Hack

Monetary damages?

All current and former T-Mobile customers were given \$25 cash payment. Customers who had to deal with identity theft were eligible for up to \$25,000 with documentation to support the claim. Lastly, all customers were offered 2 years free of charge of McAfee's ID Theft Protection Service. T-Mobile also had to put forth \$150 million to their data security.

Loss of intellectual property?

The breach resulted in a lot of compromised sensitive customer data and property owned by T-Mobile. We only know that customer data was stolen, but T-Mobile didn't disclose if other information was compromised as well.

Works Cited

<https://krebsonsecurity.com/2021/08/t-mobile-investigating-claims-of-massive-data-breach/>

https://www.cisa.gov/sites/default/files/publications/CISA_Fact_Sheet_Protecting_Sensitive_and_Personal_Information_from_Ransomware-Caused_Data_Breaches-508C.pdf

<https://www.idstrong.com/sentinel/the-saga-of-t-mobile-data-breach/>

<https://www.cnet.com/personal-finance/deadline-passes-on-t-mobiles-350-million-settlement-days-after-another-data-breach/>

[T-Mobile suffers second data breach this year - Help Net Security](#)